# ABHAY MISHRA

**Computational Engineering** | **Applied Mathematics** | **Software Engineering & DevOps**

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**mishra.se** | **LinkedIn** | **GitHub**

## PROFILE

Engineering Physics student at Uppsala University specializing in **Applied Computational Engineering**, combining **applied mathematics**, **numerical analysis**, **scientific computing**, **machine learning** and **computational modelling** with professional experience in **software engineering**, **DevOps** and **cloud-native systems**. Experienced in **distributed backend services**, **microservices**, **Kubernetes**, **CI/CD**, **Linux**, **automated testing**, **observability**, **data engineering**, **computer vision** and **image-analysis workflows** for aerospace and Earth-observation applications.
Completed upper-secondary education **two years ahead of schedule** and currently following an **accelerated university study plan**, targeting graduation in **2028 — one year ahead of the standard Engineering Physics programme timeline**.

## PROFESSIONAL EXPERIENCE

**Junior Software Engineer – Space Computing & Cloud Systems** **Oct 2025 – Aug 2026**

*BruhnBruhn Innovation* *Uppsala, Sweden*

***Python** · **Linux** · **Microservices** · **Kubernetes** · **GitLab CI/CD** · **DevOps** · **Observability** · **Machine Learning** · **Computer Vision** · **Image Analysis***

- Developed and maintained **containerized microservices** and **backend services** for cloud-native aerospace and **Earth-observation** software platforms.
- Supported **Kubernetes deployments**, **GitLab CI/CD**, **container orchestration** and **software-lifecycle workflows** across distributed computing environments.
- Developed platform **observability**, **heartbeat monitoring** and **system-status services**, and contributed to **automated testing** and **validation infrastructure**.
- Developed and evaluated **machine-learning** and **computer-vision** workflows for Earth-observation and real-time imagery, including **image analysis**, **object detection**, **classification**, **GPU-based model training** and **model-validation pipelines**.
- Worked across **backend implementation**, **platform integration**, **deployment infrastructure**, **data processing** and operational validation.

**Summer Research Intern – Materials Theory** **Jun 2025 – Sep 2025**

*Uppsala University, Department of Physics and Astronomy* *Uppsala, Sweden*

***Python** · **Numerical Analysis** · **Computational Physics** · **Applied Linear Algebra** · **Scientific Computing***

- Developed **computational software** for determining longitudinal spin susceptibility from materials-theory **simulation data**.
- Implemented **numerical solution methods** for coupled Dyson-equation formulations using static bare and interacting susceptibility data derived from LSDA/DFT calculations.
- Built reusable **numerical-processing**, **data-analysis** and **visualization pipelines** for simulation outputs and research validation.

## SELECTED ENGINEERING PROJECTS

**mishra.se – Personal Engineering Portfolio** **2026 – Present**

***Web Development** · **Technical Portfolio** · **Engineering Communication***

- Designed and maintain a personal engineering portfolio presenting **professional software experience**, **computational projects** and technical engineering work in a recruiter-facing format.
- Structured the platform around the intersection of **computational engineering**, **software development**, **machine learning** and **applied mathematics**.

**RoboticHandControl – Embedded Bionic Hand** **2023 – 2024**

***C++** · **Arduino** · **Embedded Control** · **Servo Actuation** · **Wireless Communication***

- Designed and programmed a remotely controlled **bionic hand**, integrating mechanical design, **embedded control**, servo actuation and wireless communication.
- Implemented **hardware–software integration** and **real-time actuator control**; awarded **Best Practical Graduation Project**.

**DataComparisonBio – Bioinformatics Data Analysis** **2024**

***Python** · **Data Analysis** · **Statistical Processing** · **Visualization***

- Developed **Python** tooling for genomic dataset comparison, **statistical analysis** and reproducible **data visualization**.
- Automated recurring **data-processing workflows** to improve consistency and reproducibility across analyses.

**GraphCalculator – Numerical Computing Application** **2023**

***Python** · **NumPy** · **Matplotlib** · **Tkinter***

- Developed an interactive **numerical computing** application supporting function plotting, **numerical differentiation** and intersection analysis through a desktop GUI.

## EDUCATION

**M.Sc. in Engineering – Engineering Physics** **Aug 2024 – Jun 2028**

*Uppsala University* *Uppsala, Sweden*

- Specialization: **Applied Computational Engineering**, with focus on **applied mathematics**, **numerical methods**, **computational modelling**, **scientific computing** and **high-performance computing**.
- Following an **accelerated academic plan** with a substantially above-standard course load and planned graduation approximately **one year ahead of the standard five-year programme**.

**Technology Programme – Information & Media Technology** **Aug 2021 – Jun 2024**

*NTI Gymnasiet* *Uppsala, Sweden*

- Completed upper-secondary studies approximately **two years ahead of the conventional academic timeline**.
- Awarded **Best Practical Graduation Project** for the RoboticHandControl embedded engineering project.

## TECHNICAL SKILLS

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|---|---|
| **Programming & Systems** | **Python**, **C++**, **C#**, **JavaScript**, **MATLAB**, **Linux**, **Git** |
| **Machine Learning & Vision** | **Machine Learning**, **Computer Vision**, **Image Analysis**, **Object Detection**, **Classification**, **Model Evaluation**, **GPU-Based Model Training**, **Real-Time Video Analytics** |
| **Data & Numerical Computing** | **Data Analysis**, **Data Engineering**, **Numerical Analysis**, **Scientific Computing**, **NumPy**, **SciPy**, **Matplotlib**, **Simulation Data Processing**, **Numerical Visualization** |
| **Software Engineering & DevOps** | **Backend Development**, **Microservices**, **Distributed Services**, **Containerized Applications**, **Kubernetes**, **GitLab CI/CD**, **Automated Testing**, **Observability**, **Software Lifecycle Workflows** |
| **Computational Engineering** | **Mathematical Modelling**, **Numerical Methods**, **Computational Physics**, **Simulation Analysis**, **Applied Linear Algebra**, **Scientific & Engineering Computation** |
| **Web & Engineering Tools** | **Vue.js**, **React**, **HTML/CSS**, **Flutter**, **Figma**, **LaTeX** |

## LEADERSHIP & ENGAGEMENT

**Vice President, F1RST Uppsala** **Jul 2026 – Present**
Board coordination, executive decision support and **organizational development**. Uppsala

**Marketing & Public Relations Manager, UTNARM 2026** **Mar 2026 – Present**
Lead **marketing**, **corporate engagement** and student outreach for Uppsala University's engineering and science career fair. Uppsala

**Convening Representative, UTN Council (FUM)** **Jan 2026 – Dec 2026**
**Student-union governance**, representation and organizational decision-making. Uppsala

**Course & Safety Representative, Engineering Physics** **Aug 2024 – Sep 2027**
**Student representation** concerning **education quality**, course development and study environment. Uppsala

## ADDITIONAL EXPERIENCE

**Student Ambassador – Faculty of Engineering and Science, Uppsala University** **Oct 2025 – Present**

**Long-Term Substitute Teacher – Mathematics & Physics, NTI Gymnasiet** **Aug 2024 – Jun 2025**

**Academic Coach – Allakando** **Jun 2024 – Oct 2024**

**Field Assistant & Statistical Analytics Intern – SLU** **Jun 2024 – Jul 2024**

## LANGUAGES

**Swedish** – Fluent **English** – Fluent **German** – A2 **Hindi** – Native spoken